

GMM 1&2



Section 1: A discussion of the data as presented.

Define the **Data & Samples** in this exercise.

Identify the mistakes in the data. **Mistakes** can be corrected.

Errors are part of the data (dispersion).

Section 2: What parameter(s) give you confidence about how representative of the actual samples (real) the calculated modulus is?

At this point we are still talking about individual datapoint, what we are interested in is **each** calculated Young's modulus.

Section 3: What's the **error** in the calculated modulus? Correctly expressed the moduli with its **significant figures**. What are the **units** of the calculation in the original table? What units would be best to use?

How did you calculate the modulus?

Which figure is certain, which figure starts to be uncertain (rounded).

Data treatment



Single measurement
Strain/stress

- Identify the mistake – if you can resolve – resolve it.
- Identify the mistake – if you cannot resolve – delete it.
- Can not identify the mistake – try to calculate and compare whether there is significant change on the sample mean with and without this data point.
- ALWAYS avoid deleting data, unless you REALLY know there is a mistake and you cannot resolve it.

Slope

Repeated measurements
Young's modulus (Data)

Mean, SD, SE, etc.

Samples:

Unmodified materials
Unmodified polymers
Modified materials
Modified polymers

- Data do NOT have a single value
- Units, certain & uncertain figures

Mean $\pm$ **St Dev**

Mean $\pm$ **x% CI**

Mean + **CV**

Significant figures +/- corresponding error

Location $\pm$ spread

Significance Figures



Young's modulus = **9**67491.8332Pa

(6 would be the first figure affected by error, in **10kPa**)

STEYX → SE of Y = 9035.967904Pa (Here we use SE of Y to represent SE of slope)

CI. (95% confidence) = 1.96 * SE = **1**7710.49709Pa

(shows the order of **magnitude of the error, 10kPa**)

So, if we rewrite the both the value and error in kPa

Value → **Certain figures (9)** plus **the first uncertain figure** (6 rounded as **7**),
and finish with a 0 in the kPa place → **970kPa**.

Error → we should keep **the first uncertain figure** (1 rounded as **2**),
and finish with a 0 in the kPa place → **20kPa**.

$$970 \pm 20kPa$$

$$0.97 \pm 0.02MPa$$